DEPARTMENT OF MATHEMATICS, I.I.T. MADRAS

MA 1101 - Calculus

Assignment - 1

(1) Find the limit, if exists:

(a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{e^y \sin x}{x}$ (b) $\lim\limits_{\substack{(x,y)\to(0,0)\\ x\neq y}} \dfrac{x-y+2\sqrt{x}-2\sqrt{y}}{\sqrt{x}-\sqrt{y}}$

(c) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy}{|xy|}$ (d) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2}{x^2-y}$

(e) $\lim\limits_{(x,y)\to(0,0)} \cos\left(\dfrac{x^3-y^3}{x^2+y^2}\right)$ (f) $\lim\limits_{(x,y)\to(0,0)} \dfrac{2x}{x^2+x+y^2}$

(2) At what points (x,y) or (x,y,z) are the functions below continuous?

(a) $g(x,y)=\dfrac{x^2+y^2}{x^2-3x+2}$ (b) $h(x,y,z)=\dfrac{1}{x^2+z^2-1}$.

(3) Let $f(x,y)=\ln\left(\dfrac{3x^2-x^2y^2+3y^2}{x^2+y^2}\right)$ for $(x,y)\neq(0,0)$. Define $f(0,0)$ in a way that extends f to be continuous at the origin.

(4) Let $f(x,y)=\dfrac{y}{x^2+1}$. Given $\epsilon=0.05$, find a $\delta>0$ such that for all (x,y) with $\sqrt{x^2+y^2}<\delta$, we have $|f(x,y)-f(0,0)|<\epsilon$.

(5) Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$:

(a) $f(x,y)=\dfrac{x}{x^2+y^2}$ (b) $f(x,y)=e^{xy}\ln y$.

(6) Find f_x, f_y and f_z:

(a) $f(x,y,z)=\sin^{-1}(xyz)$ (b) $f(x,y,z)=yz\ln(xy)$.

(7) Does a function f with continuous first partial derivatives throughout an open region R have to be continuous on R?

(8) Find all the second order partial derivatives:

(a) $g(x,y)=x^2y+\cos y+y\sin x$ (b) $g(x,y)=\tan^{-1}(y/x)$.

(9) Express $\partial z/\partial u$ and $\partial z/\partial v$ in terms of u and v, where $z=4e^x\ln y,\ x=\ln(u\cos v),\ y=u\sin v$.

(10) Express $\partial w/\partial u$ and $\partial w/\partial v$ in terms of u and v, where $w=xy+yz+zx,\ x=u+v,\ y=u-v,\ z=uv$.

(11) Find dy/dx if $xe^y+\sin xy+y-\ln 2=0$.

(12) If $z=x^2+y^2+z^2$ and $z=x^2+y^2$, then find

(a) w_y, with y and z as independent variables.

(b) w_z, with z and x as independent variables.

(c) w_z, with y and z as independent variables.

(13) Suppose that $f(x,y,z,w)=0$ and $g(x,y,z,w)=0$ determine z and w as differentiable functions of the independent variables x and y. Suppose also that $f_zg_w-f_wg_z\neq 0$. Then show the following:

(a) $\dfrac{\partial z}{\partial x}=-\dfrac{f_xg_w-f_wg_x}{f_zg_w-f_wg_z}$ (b) $\dfrac{\partial w}{\partial y}=-\dfrac{f_zg_y-f_yg_z}{f_zg_w-f_wg_z}$.

(14) Find the directional derivative of f at P_0 in the direction of $\vec{A}$:

(a) $f(x,y)=\tan^{-1}(y/x)+\sqrt{3}\sin^{-1}(xy/2),\ P_0(1,1),\ \vec{A}=3\hat{i}-2\hat{j}$

(b) $f(x,y,z)=x^2+2y^2-3z^2,\ P_0(1,1,1),\ \vec{A}=\hat{i}+\hat{j}+\hat{k}$.

(15) By about how much will $g(x,y,z) = x + x\cos z - y\sin z + y$ change if the point P moves from $(2,-1,0)$ a distance $ds = 0.2$ units toward the point $(0,1,2)$?

(16) Find the directions in which the functions increase and decrease most rapidly at P_0. Then find the derivative of the function in these directions:
(a) $f(x,y) = x^2 + xy + y^2,\ P_0(-1,1)$ (b) $f(x,y,z) = xe^y + z^2, P_0(1,\ln 2, 1/2)$.

(17) Find the equation of the tangent plane and normal line at P_0 on the given surface.
(a) $x^2 + y^2 - z^2 = 18,\ P_0(3,5,-4)$. (b) $x^2 - xy - y^2 - z = 0,\ P_0(1,1,-1)$.

(18) Use Taylor's formula for $f(x,y) = (1 - x - y + xy)^{-1}$ at the origin to find quadratic and cubic approximations of f near the origin.

(19) Use Taylor's formula to find a quadratic approximation of $e^x \sin y$ at the origin. Estimate the error in the approximation if $|x| \le 0.1$ and $|y| \le 0.1$.

(20) Find all the local maxima, local minima, and saddle points of the following functions:
(a) $f(x,y) = x^3 + y^3 + 3x^2 - 3y^2 - 8$ (b) $f(x,y) = 2x^3 + 2y^3 + -9x^2 + 3y^2 - 12y$
(c) $f(x,y) = y\sin x$ (d) $f(x,y) = e^{2x}\cos y$

(21) Find the absolute maxima and minima of the functions on the given domains:
(a) $f(x,y) = 2x^2 - 4x + y^2 - 4y + 1$ on the closed triangular plate bounded by the lines $x = 0,\ y = 2,\ y = 2x$ in the first quadrant.
(b) $f(x,y) = (4x - x^2)\cos y$ on the rectangular plate $1 \le x \le 3,\ -\pi/4 \le y \le \pi/4$.

(22) Find $a, b \in \mathbb{R}$ with $a \le b$ such that I has its largest value:
(a) $I = \int_a^b (6 - x - x^2)\,dx$ (b) $I = \int_a^b (24 - 2x - x^2)^{3/2}\,dx$

(23) Find the maxima, minima, and saddle points of (x,y), if any, given that $f_x = 9x^2 - 9$ and $f_y = 2y + 4$.

(24) For what values of the constant k does the Second Derivative Test guarantee that the function $f(x,y) = x^2 + kxy + y^2$ will have a saddle point at $(0,0)$? A local minimum at $(0,0)$? For what values of k is the Second Derivative Test inconclusive? Give reasons for your answers.

(25) Find the absolute maximum and minimum values of $g(x,y) = 2x^2 + y^2$ on the quarter circle $x^2 + y^2 = 4,\ x \ge 0,\ y \ge 0$.

(26) Find the points on the curve $x^2 + xy + y^2 = 1$ in the xy-plane that are nearest to and farthest from the origin.

(27) The temperature at a point (x,y) on a metal plate is $T(x,y) = 4x^2 - 4xy + y^2$. An ant on the plate walks around the circle of radius 5 centered at the origin. What are the highest and lowest temperatures encountered by the ant?

(28) A space probe in the shape of the ellipsoid $4x^2 + y^2 + 4z^2 = 16$ enters Earth's atmosphere and its surface begins to heat. After 1 hour, the temperature at the point (x,y,z) on the probe's surface is $T(x,y,z) = 8x^2 + 4yz - 16z + 600$. Find the hottest point on the probe's surface.

(29) Find the extreme values of the function $f(x,y,z) = xy + z^2$ on the circle in which the plane $y - x = 0$ intersects the sphere $x^2 + y^2 + z^2 = 4$.

Answers:

1a. 1 1b. 2 1c. -1 1d. doesnt exist 1e. 1 1f. doesnt exist 2a. $x \ne 1, x \ne 2$
2b. $x^2 + z^2 \ne 1$ 3. $f(0,0) = \ln 3$ 4. $\delta = 0.05$ 5a. $-\frac{2xy}{(x^2+y^2)^2}$ 5b. $xe^{xy}\ln y + \frac{e^{xy}}{y}$ 7. Yes

14a. $-3/(2\sqrt{13})$ 14b. 0 15. 0 16. Increases most in direction (a) $-\hat{\imath}/\sqrt{2}+\hat{\jmath}/\sqrt{2}$ (b) $(2\hat{\imath}+2\hat{\jmath}+k)/3$ 17. Tangents: (a) $3x+5y+4z=18$ (b) $x-3y-z=-1$ 18. Cubic Approximation: $1+x+y+x^2+xy+y^2+x^3+x^2y+xy^2+y^3$ 19. Quad Approx: $y+xy$, Error: ≤ 0.000814 20a-b. one loc min, one loc max, two sadl pts. 20c. only sadl pts 20d. no extrema, no sadl pts 21a. Max: 1, Min: -5 21b. Max: 4, Min: $3/\sqrt{2}$ 22a. $a=-3,\ b=2$ 22b. $a=-6,\ b=4$ 23. Loc Min at $(1,-2)$, no loc. max, sadl pt at $(-1,-2)$ 24. Loc min for $|k|<2$, Sadl pt for $|k|>2$, Inconclusive for $|k|=2$ 25. Max. 2, Min. 0 26. Nearest: $\pm(1/\sqrt{3},1/\sqrt{3})$, Farthest: $\pm(1,-1)$ 27. Highest: 125°, Lowest: 0° 28. $(\pm 4/3,-4/3,-4/3)$ 29. Max: 4, Min: 2